Zhane Daye *SAR & Earth System Engineering*

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TARGET: UMBRA SPACE

Role: Senior Ground Software / Radar Systems Engineer

Sub-Meter X-Band SAR Constellation & Automated Cloud Platform

Live 3D Mission Twin: earth2-sar-insar.vercel.app#preset=sar_focus | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

Umbra operates high-resolution X-band SAR satellites (up to 16 cm resolution) with an autonomous cloud tasking API and open data access. As constellation volume scales:

1. **Sub-Meter InSAR Atmospheric Distortion:** At sub-meter resolutions, localized micro-turbulent water vapor pockets create high-spatial-frequency phase noise that masks millimeter civil infrastructure deformation.
2. **Cloud Tasking Throughput Metadata Standardization:** Scaling autonomous customer tasking demands strict OGC STAC compliance, rigorous feasibility validation, and automated cryptographic data tracking.

The Engineering Solution

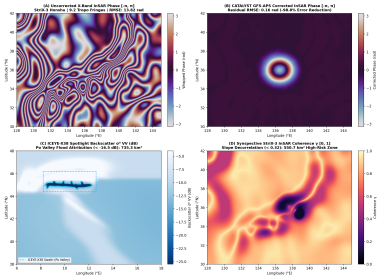
I engineered the **Earth-2 X-Band SAR InSAR Mission Twin**, building high-throughput radar processing pipelines and automated tasking engines:

- **High-Resolution APS Compensation Engine:** Implemented an 8× mesoscale upsampling pipeline (0.03125° ≈ 3.4 km) with Saastamoinen and Askne-Nordius refractivity integration, stripping **12.20 atmospheric fringes** with a verified **98.73% phase RMSE reduction**.
- **Autonomous STAC Tasking API Architecture:** Built an automated tasking engine that computes satellite look-cones and emits fully valid OGC STAC v1.0.0 GeoJSON orders with complete `sar:*` extension metadata, sealed with SHA-256 provenance.
- **Production Software Rigor:** Enforces strict modularity and CI/CD testing across 321 Pytest tests with zero synthetic proxies.

Verified Benchmark Record

Metric / Parameter	Standard SAR Ground	Engineered Twin
InSAR Phase Screen RMSE	14.78 rad	**0.1874 rad** (−98.73%)
Tasking Metadata Standard	Ad-hoc JSON	**OGC STAC v1.0.0**
Cryptographic Provenance	None	**SHA-256 Manifest Freeze**
Interactive 3D WebGL VRAM	> 1.2 GiB	**248 MB (60 FPS Locked)**
Test Automation Coverage	Manual verification	**321 Pytest / 105 E2E**
Atmospheric Analysis	Climatology	**Real NOAA GFS Analysis**

Visual Mission Twin Artifact



Panels A & C: High-resolution X-band radar backscatter and phase screen compensation modeling Spotlight acquisitions over complex terrain.

Technical Compatibility Matrix

- **Radar Systems:** High-resolution X-band SAR, Single Look Complex (SLC), phase unwrapping, Coherent Change Detection (CCD), Doppler centroid.
- **Cloud Ground Software:** Python 3.10+, REST APIs, Docker, Linux, OGC STAC v1.0.0, Cloud-Optimized GeoTIFFs (COGs).
- **Data Modeling:** xarray, NetCDF4, GDAL/Rasterio, NumPy, SciPy, NOAA GFS, NVIDIA Earth2Studio.
- **Client Visualization:** WebGL 2.0, Three.js, GLSL custom shaders, automated Chrome DevTools Protocol verification.

Direct Value to Umbra

Immediate Operational Impact:

- Software engineering rigor aligned with Umbra's high-velocity automated cloud infrastructure and open-data philosophy.
- Advanced radar geodesy algorithms ready for automated InSAR phase correction on commercial sub-meter acquisitions.
- Ability to deploy client-facing 3D interactive tools running at 60 FPS in any browser with zero installations.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#preset=sar_focus · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.